

PARTH SANJAY SAPRE

Data Engineer

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PROFESSIONAL SUMMARY

Data Engineer with 5+ years of experience building modeling solutions for healthcare challenges, insurance, and financial risk sectors. Directed production-ready Data, AI, ML, and reporting solutions, processing over 500M+ healthcare records daily and orchestrating end-to-end AWS workflows.

WORK EXPERIENCE

Tria Federal

August 2026 – Present

Data Engineer

- Develop and optimize PySpark/Spark SQL pipelines for CMS RASS using AWS EMR, Glue, and S3 to support large-scale risk-adjustment data processing.
- Build and maintain automated workflows using Lambda, Step Functions, CloudFormation, Redshift, and Snowflake for data transformation, orchestration, and downstream processing.
- Perform unit, integration, regression, and data-validation testing across RASS workflows to ensure data accuracy and processing reliability.
- Tune Spark/EMR workloads using partitioning, caching, join optimization, and resource configuration to improve scalability and stability.
- Train developers and testers on RASS business logic, PySpark, AWS architecture, debugging, and testing best practices.

Technology Square Inc [IBM Contractor]

June 2022 – August 2026

PySpark/Python Engineer (Richmond, VA)

- Collaborated and developed RASS, a CMS healthcare risk adjustment platform conceptualized to calculate beneficiary-level risk scores and generate plan-facing reports for Medicare health plans.
- Engineered a RAG (Retrieval-Augmented Generation) pipeline with LLM orchestration that ingested 1.2 thousand CMS PDF pages, extracted model logic, and cut manual review time by 120 hours monthly, uncovering three most frequent policy ambiguities.
- Automated reports, score validation files, reconciliation outputs, exception reports, and AI-assisted unit testing, regression testing, and data quality validation.
- Performed rigorous Exploratory Data Analysis, feature engineering, and deep learning model validation for text classification, achieving 99.9% accuracy in predictive outputs aligned with critical healthcare business rules.
- Processed 500M+ healthcare records, confidential information and approximately 2.8 TB/day using PySpark, Spark SQL, AWS EMR, Glue, large-scale distributed datasets and S3 for large-scale risk score calculation.
- Improved production reliability by building Splunk dashboards, workflow monitoring, validation checks, and failure alerts, reducing production failures by up to 85% and improving recovery time by approximately 70%.

Nest Technology

September 2021 – May 2022

Data Engineer (Sterling, VA)

- Generated a homeowners insurance premium prediction solution using property, claims, location, coverage, and risk data, enhancing analysis accuracy by 15%.

- Optimized machine learning models, including Linear Regression, Random Forest, and XGBoost, to achieve a 15% decrease in prediction error and a 10% uplift in underwriting accuracy.
- Spearheaded comprehensive model validation using RMSE, MAE, R-squared, and cross-validation, enhancing prediction accuracy and reducing error rates by 15%.
- Created dashboards and visual reports to present premium trends, claim impact, location-based pricing patterns, and risk segments for business stakeholders.

Gannon University

January 2020 – January 2021

Research Assistant (Erie, PA)

- Constructed data preprocessing workflows for NLP tasks, generating clean datasets that powered 12 downstream machine-learning services accessed through secure API calls.
- Initiated a secure data sharing framework for healthcare, achieving 100% FHIR compliance and processing 10,000 transactions per second for critical patient information.
- Fabricated secure Python scripts for streamlined data sharing, integrating automated testing that confirmed 99.9% data integrity and system reliability for critical applications.

PROJECTS

Glassia AI | AI Voice Generation & Localization Platform

- Architected Glassia AI, a full-stack GenAI voice platform supporting transcription, text-to-speech, multilingual voice overs, translation, voice cloning, subtitle generation and real-time lip-sync workflows enhancing user engagement by 40%.

Credit Risk Classifier | Python, Scikit-learn

- Executed a machine learning credit risk classifier using Python and Scikit-learn, predicting high-risk transactions and identifying the three biggest causes of transaction risk.
- Improved risk identification by detecting 81% of high-risk transactions through feature engineering, model training, validation, and performance evaluation, helping reduce potential financial losses.

EDUCATION

- **Gannon University**, Erie, PA 2021
Computer and Information Science (Data Science)
Award: Scholastic Excellence Award
- **Dharmsinh Desai University**, Nadiad, INDIA 2019
Computer Engineering (B. Tech)

SKILLS

- **Core AI/ML Technologies:** Generative AI, RAG, LLMs, Prompt Engineering, Fine-Tuning LLMs, measuring/benchmarking and deploying LLMs, Vector Embeddings, Semantic Search, LangChain, Transformers, Hugging Face, OpenAI, BERT, Claude, LLAMA/Mistral
- **Machine Learning Frameworks & Libraries:** TensorFlow, PyTorch, Keras, Scikit-learn, NLTP, Pandas, NumPy, XGBoost, Gradient Boosting, Logistic Regression, Random Forest
- **Data Processing & Platforms:** PySpark, Spark SQL, Apache Spark, ETL Pipelines, Data Warehousing, Batch Processing, Data Quality Checks, Data Lakes, Distributed Computing
- **Cloud Infrastructure & Deployment:** AWS, GCP, EMR, Glue, Lambda, S3, Redshift, BigQuery, CloudWatch, Docker, CI/CD, Git, Jenkins, GitHub, Git, BitBucket
- **Agile Environment:** Agile methodologies, Scrum, Kanban

CERTIFICATIONS

Machine Learning Specialization – Coursera (Andrew Ng), Google Machine Learning, Databricks Data Engineer, Certified AWS Solutions Architect, Certified AWS Cloud Practitioner, PySpark, Python, Natural Language Processing, Machine Learning, Pattern Recognition, Advanced SQL, AWS.